Multiple-choice questions

|!|BCEMB0110000001|!| ★

The following photographs show the leaves of two kinds of plants, P and Q.



Leaves of plant P



Leaves of plant Q

Which of the following ***most*** probably shows the stomatal densities (number/mm^2) in the lower epidermis of the leaves of plants P and Q?

	Plant P	***Plant Q***
A.	0	0
B.	0	35
C.	12	0
D.	12	35

□

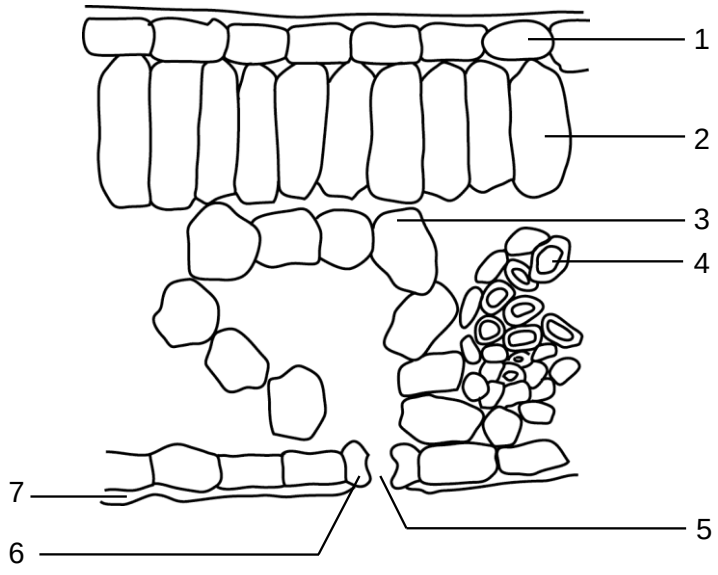
##B##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0110000002|!| ★

The diagram below shows a transverse section of a terrestrial dicotyledonous leaf.



Which of the following structures is/are absent in the submerged leaves of an aquatic plant?

(1) structure 4
(2) structure 6
(3) structure 7

A. (2) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0110000003|!| ★

Which of the following combinations correctly state the feature and the adaptation of trees in tropical rainforests?

	Feature	***Adaptation***
(1)	tall tree	outcompetes other trees to obtain more sunlight
(2)	broad leaves	increases the surface area of leaves to receive sunlight
(3)	thick cuticle on leaf surface	reduces water loss

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

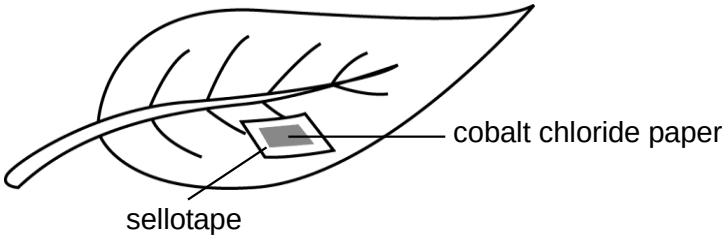
□

##A##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0110000005|!| ★ **SI**

In an investigation, a piece of dry cobalt chloride paper was attached by sellotape to both the upper and lower surfaces of a freshly picked green leaf as shown below.



After 10 minutes, the cobalt chloride paper on the upper surface remains blue whereas that on the lower surface has turned pink. Which of the following is the ***best*** explanation for this difference?

A. More guard cells are present on the upper surface.
B. More guard cells are present on the lower surface.
C. The upper surface has a higher rate of transpiration.
D. The lower surface has a higher rate of transpiration.

□

##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0110000006|!|

Which of the following sets of environmental conditions would likely lead to the highest rate of transpiration?

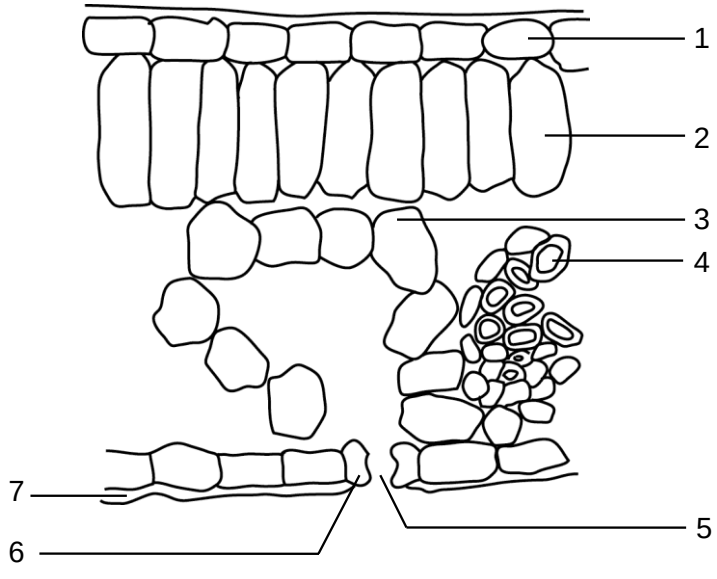
A. low humidity and low temperature
B. low humidity and high temperature
C. high humidity and high temperature
D. high humidity and low temperature

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0110000007|!|

The diagram below shows a transverse section of a terrestrial dicotyledonous leaf.



Which of the following structures help(s) regulate water loss from the leaf?

A. structure 5 only
B. structure 6 only
C. structures 5 and 7 only
D. structures 5, 6 and 7 only

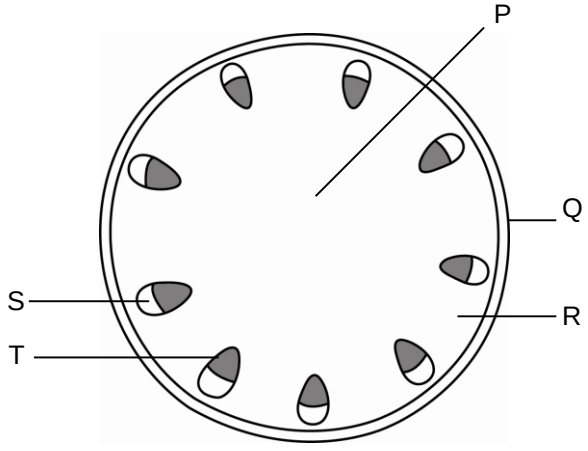
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##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0110000008|!|

The diagram below shows a transverse section of a young dicotyledonous stem.



Where does the transport of amino acids occur in the stem?

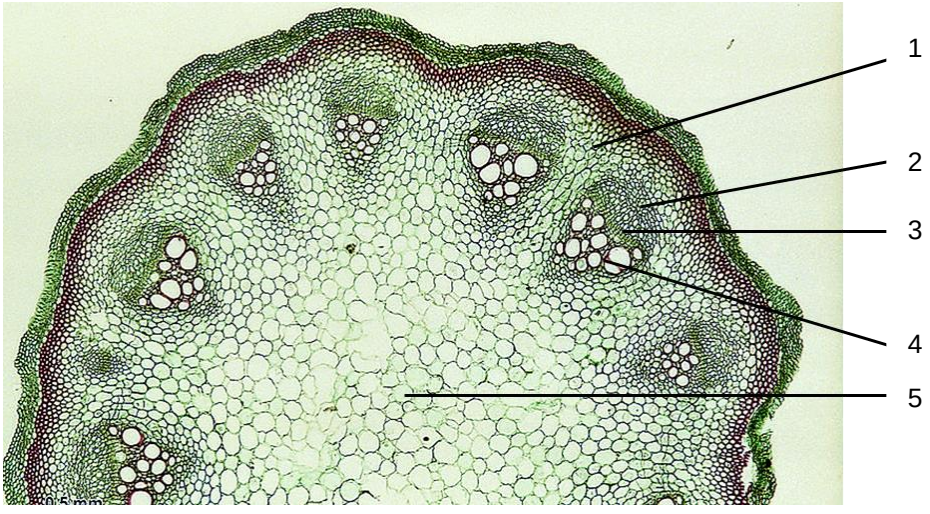
A. P
B. Q
C. S
D. T

##C##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0110000009|!| ★

The photomicrograph below shows a transverse section of part of a young dicotyledonous stem.



(Source: Doc. RNDr. Josef Reischig, CSc.)

If an aphid is going to insert its mouthparts into the stem, which part of the stem should the mouthparts reach?

A. 1
B. 3
C. 4
D. 5

##B##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0110000010|!|

A potted plant is placed under bright sunlight for several hours. At some point, its stomata close. Which of the following statements best accounts for this?

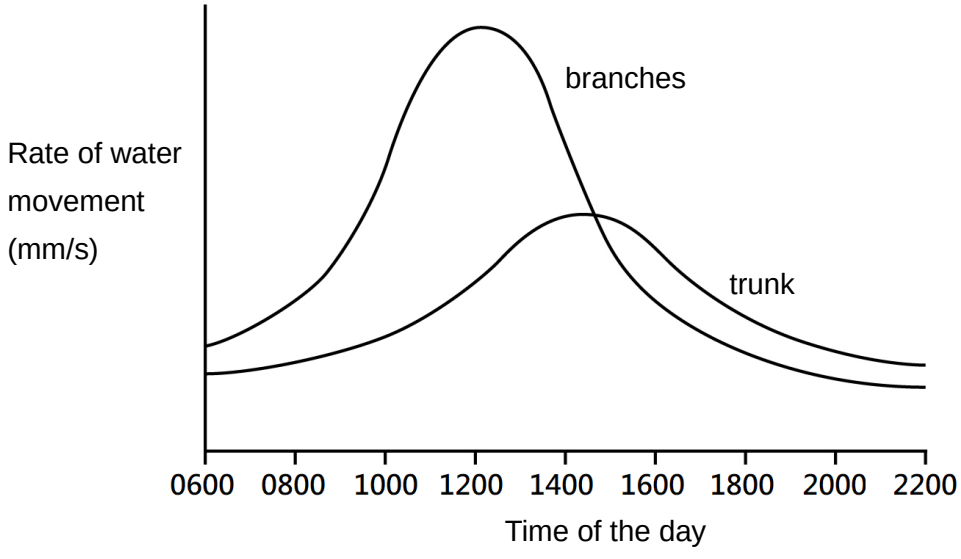
A. The photosynthetic rate of the guard cells is too high.
B. The plant has absorbed enough carbon dioxide from the air for photosynthesis.
C. The plant has lost excess water.
D. The plant does not need oxygen from the air for respiration.

##C##

Skill assessed	Mark
Applying concepts	1

|!|BCEMA0110000011|!| ★★

The graph below shows the change of the rate of water movement in the branches and the trunk of a tree during the course of a day.



It can be concluded that during the course of the day

A. the water in the branches is transported from the trunk.
B. the branches are at a higher position than the trunk.
C. the greatest increase in the water flow rate first occurs in the branches.
D. the greatest increase in the water flow rate first occurs in the trunk.

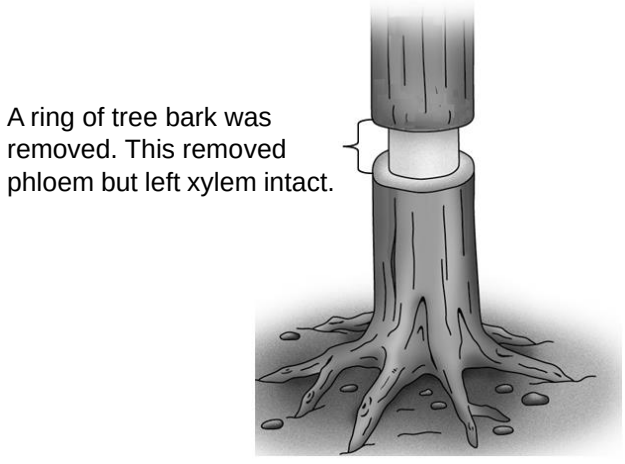
##C##

Skill assessed	Mark
Analysing data	1

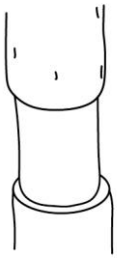
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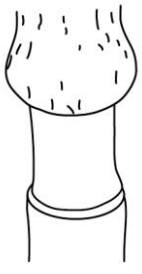
|!|BCEMC0110000012|!|

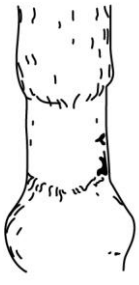
A ring of tree bark containing the phloem is removed from a healthy woody stem, leaving the xylem intact.



Which of the following correctly shows the appearance of the stem after several weeks?

A. 

B. 

C. 

D. 

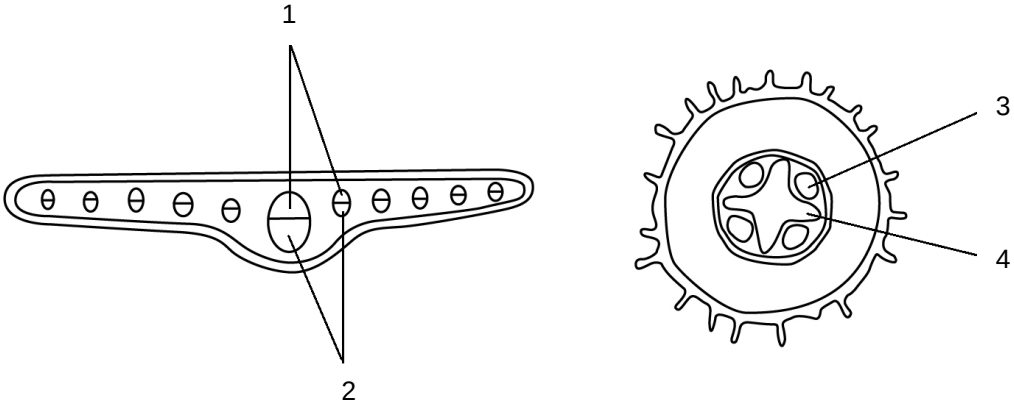
□

##B##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0110000013|!| ★

The diagrams below show the transverse sections of a leaf and a root of a dicotyledonous plant.



Leaf ***Root***

Which labelled structures transport sucrose?

A. 1 and 3
B. 1 and 4
C. 2 and 3
D. 2 and 4

##C##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0110000014|!|

An increase in factor X would reduce the rate of transpiration in plants. What is factor X?

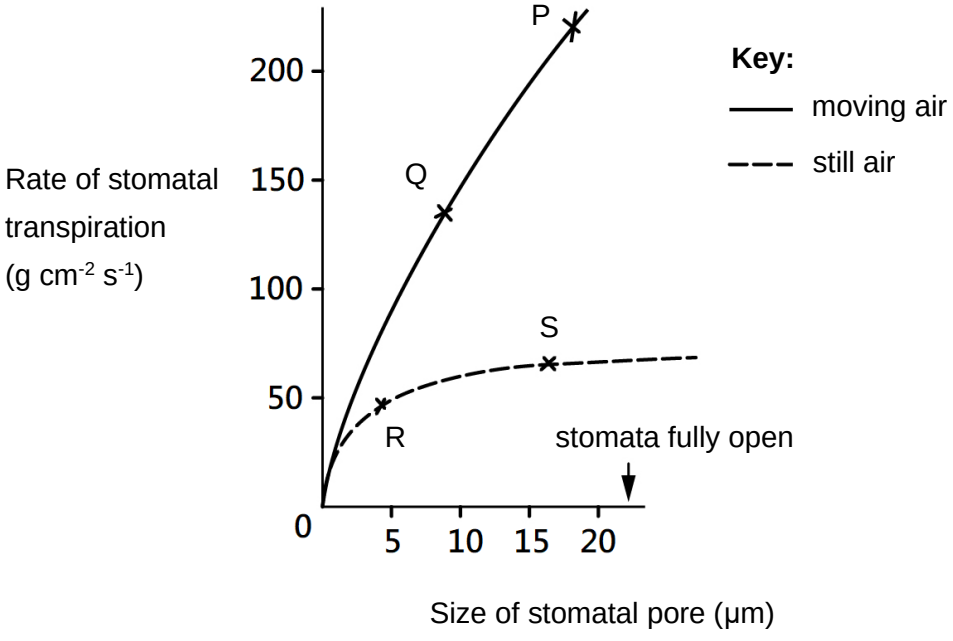
A. air movement
B. temperature
C. light intensity
D. humidity

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0110000015|!| ★

The graph below shows the relationship between the diameter of stomatal pore and the rate of transpiration in still air and in moving air.



At which point on the graph is the rate of transpiration independent of the size of the stomatal pore?

A. P
B. Q
C. R
D. S

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0110000016|!| ★

The table below shows the distribution of stomata on the upper and lower epidermis of leaves of four plant species (P, Q, R and S).

	Number of stomata per cm^2	
	Upper epidermis	**Lower epidermis**
P	2600	2250
Q	7600	0
R	0	24,300
S	9	5

Which species is most likely a plant with floating leaves?

A. P
B. Q
C. R
D. S

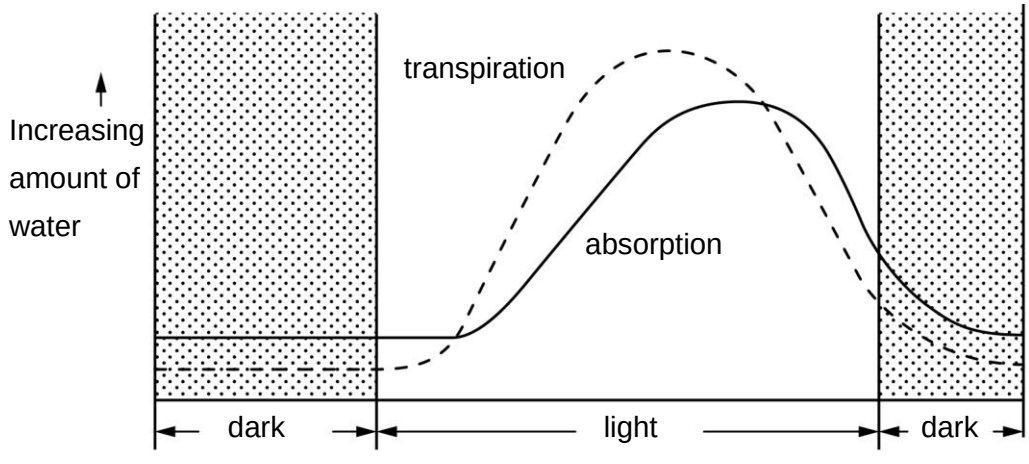
□

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0110000017|!| ★

The graph below shows the rates of transpiration and water absorption of a sunflower plant during a 24-hour period.



With reference to the graph, it can be concluded that

A. under light conditions, the water absorption rate of the plant always exceeds transpiration rate.
B. under dark conditions, the water absorption rate of the plant always exceeds transpiration rate.
C. under light conditions, the transpiration rate of the plant always exceeds the water absorption rate.
D. under dark conditions, the transpiration rate of the plant always exceeds the water absorption rate.

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0110000018|!| ★

A student put a leaf into a beaker of hot water. She noticed that fewer bubbles were given off from the upper side of the leaf than the lower side. Which of the following is a correct explanation for this?

A. Hot water can make the air in the air spaces of spongy mesophyll expand.
B. The upper epidermis has fewer stomata than the lower epidermis.
C. The waxy cuticle on the upper epidermis is thicker than that on the lower epidermis.
D. The fewer stomata on the upper epidermis help prevent excess water loss from the plant.

##B##

Skill assessed	Mark
Analysing data	1

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|!|BCEMC0110000019|!|

Which of the following statements explain(s) why a stem is both cut and connected to a potometer under water?

(1) To prevent air from entering the xylem vessels.
(2) To prevent plasmolysis of the xylem vessels.
(3) To prevent the collapse of xylem vessels.

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

##A##

Skill assessed	Mark
Practical skills	1

|!|BCEMC0110000020|!|

Which of the following substances is/are transported by xylem vessels in terrestrial plants?

(1) water
(2) minerals
(3) oxygen

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

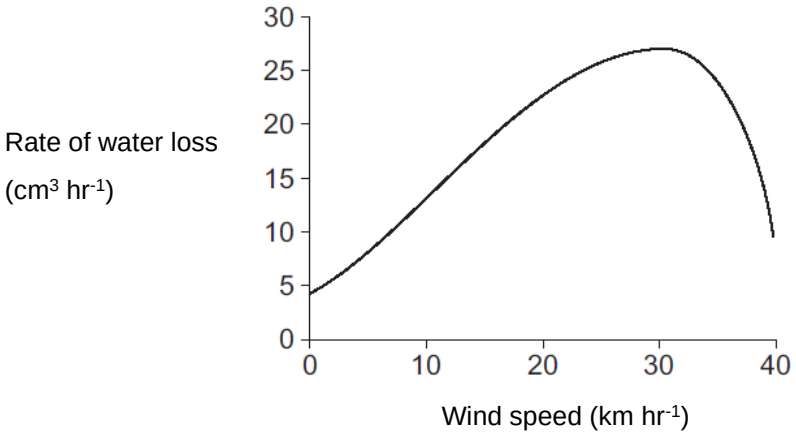
##B##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMA0110000021|!| ★★

The graph shows the results of using a weight potometer to measure the effect of wind speed on the rate of water loss in a plant.



Which of the following statements correctly accounts for the decrease in transpiration rate from 30 to 40 km/hr?

A. At high wind speeds, moist air around the stomata is removed and replaced by dry air.
B. At high wind speeds, most carbon dioxide around the stomata is blown away and the stomata need to be fully open for gas exchange.
C. At high wind speeds, the transpiration rate is high due to an increased diffusion gradient. The stomata close to prevent further water loss.
D. The rate of photosynthesis is equal to that of respiration and there is enough carbon dioxide inside the air spaces. Hence, the stomata close.

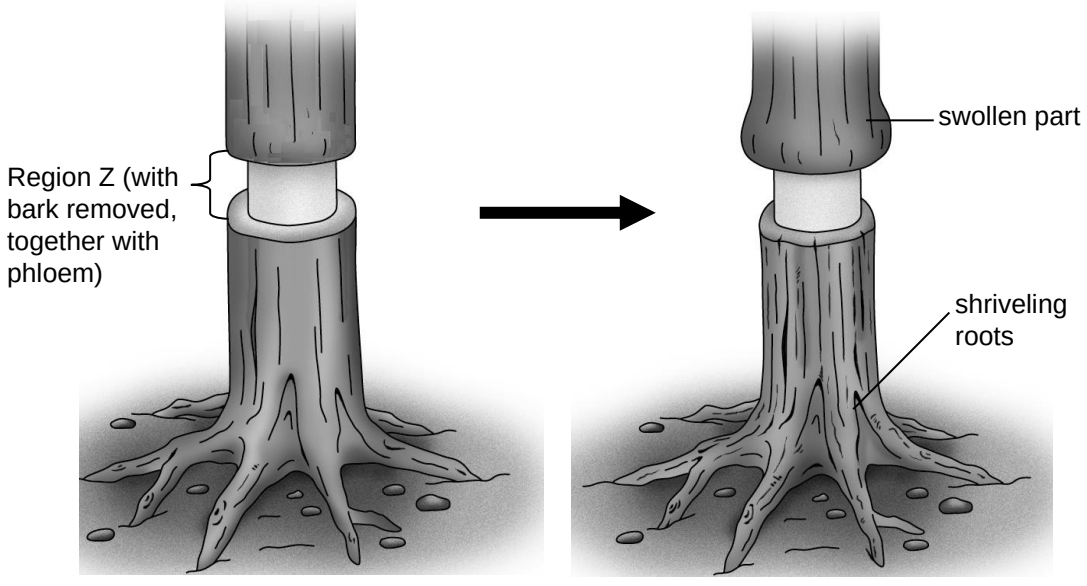
□

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0110000023|!| ★

The following diagram shows the bark ringing experiment performed by a scientist called Marcello Malpighi to study the transport in plant. The outer ring of the bark, including the phloem of a tree was first removed.



Which of the following explains the results?

A. Food is accumulated above region Z.

B. Water is accumulated within the xylem above region Z.

C. The removal of bark stimulates the growth of the xylem above region Z.

D. The removal of bark causes fungal infection of the stem above region Z.

##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0110000024|!|

In a plant, water can be lost through

(1) stomata.
(2) cuticle.
(3) lenticels.

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0110000025|!| ★

In which of the following conditions will the rate of transpiration of a plant be the highest?

A. a hot summer noon
B. a foggy morning in spring
C. inside a crop-cultivating greenhouse
D. a cloudless evening in winter

##A##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0110000026|!|

Which of the following statements about the structures of phloem is ***incorrect***?

A. Sieve tubes have no nuclei.
B. Sieve tubes have lignified cell walls.
C. Sieve plates have pores to allow organic nutrients to pass through.
D. Each companion cell contains a nucleus.

##B##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0110000028|!|

Which of the following statements about the vascular bundles is ***incorrect***?

A. The vascular bundles are found at the periphery of the root.
B. The vascular bundles can provide support to woody plants.
C. In the vascular bundles of the leaves, the xylem lies above the phloem.
D. In the stem, the xylem is located in the inner region of each vascular bundle, and the phloem in the outer region.

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0110000030|!| ★

In an experiment, the gas given off from the cut stem of an aquatic plant (e.g. *Hydrilla*) was analysed at different times of a day, with the temperature kept constant throughout the experiment. The results are shown in the following table.

Time	Weather	Percentage of carbon dioxide in the gas (%)	Percentage of oxygen in the gas (%)
7 a.m.	dull, cloudy	5	15
11 a.m.	slightly cloudy	0	30
2 p.m.	sunny	0	37
4 p.m.	sunny	0	43

Which of the following correctly explains the results at 7 a.m.?

A. Lower light intensity decreases the rate of transpiration.
B. Lower light intensity decreases the rate of respiration.
C. The rate of respiration is higher than the rate of photosynthesis.
D. The rate of transpiration is lower than the rate of photosynthesis.

##C##

Skill assessed	Mark
Analysing data	1

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|!|BCEMC0110000031|!|

Which of the following is ***not*** an adaptation of a cactus to a dry environment?

A. extensive root system
B. epidermis covered by a thick waxy cuticle
C. sunken stomata in leaves
D. needle-shaped leaves

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0110000032|!|

Which of the following descriptions about xylem are correct?

(1) It transports water and minerals in one direction.
(2) The end walls of the adjacent xylem cells are broken down.
(3) The main substance that gives rigidity to the xylem is cellulose.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

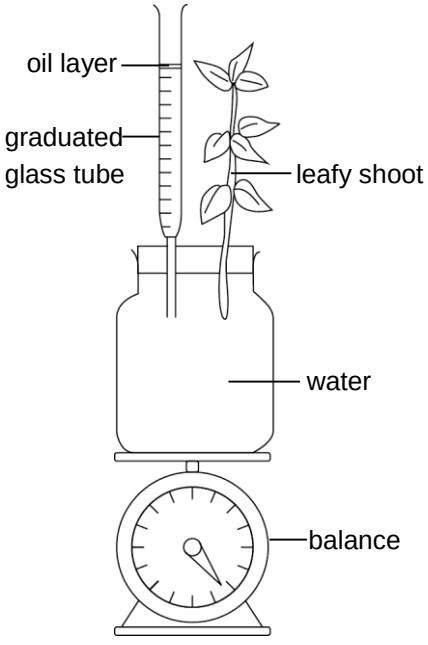
##A##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0110000033|!| ★ SI

The diagram below shows a weight potometer which was used to study the water balance of a leafy shoot.



The main function of adding a thin layer of oil in the graduated glass tube was to

A. cut off the supply of oxygen to the leafy shoot.
B. ensure the glass tube is water-tight.
C. prevent the leafy shoot from contamination.
D. prevent the evaporation of the water.

##D##

Skill assessed	Mark
Practical skills	1

|!|BCEMC0110000034|!|

Which of the following statements about transpiration is/are correct?

(1) It enables the transport of minerals from the roots to the leaves.
(2) It occurs only in the leaves.
(3) It causes the absorption of minerals from the soil into roots by active transport.

A. (1) only
B. (2) only
C. (1) and (2) only
D. (2) and (3) only

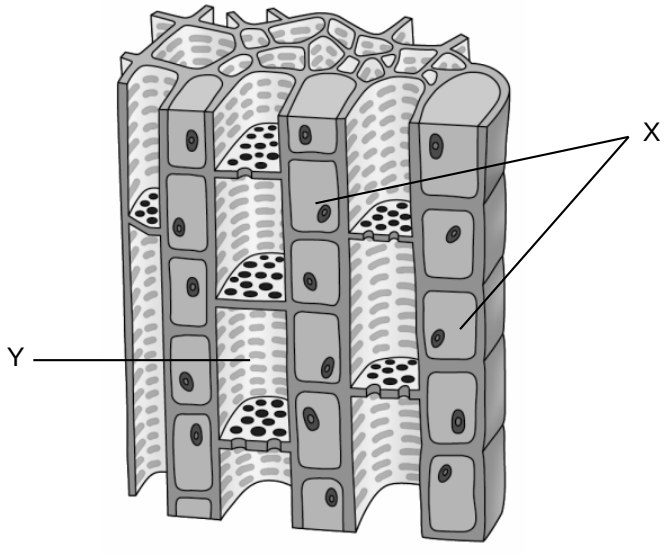
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##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0110000035|!|

The diagram below shows a part of the transport system of a plant.



Which of the following comparisons between X and Y are correct?

	X	***Y***
(1)	with cytoplasm	without cytoplasm
(2)	with mitochondria	without mitochondria
(3)	with nucleus	without nucleus

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0110000036|!|

Transpiration in plants is necessary because it

(1) helps the transport of water and minerals to the leaves.
(2) prevents the plants from becoming overheated during a hot day.
(3) helps the plants remove metabolic wastes.

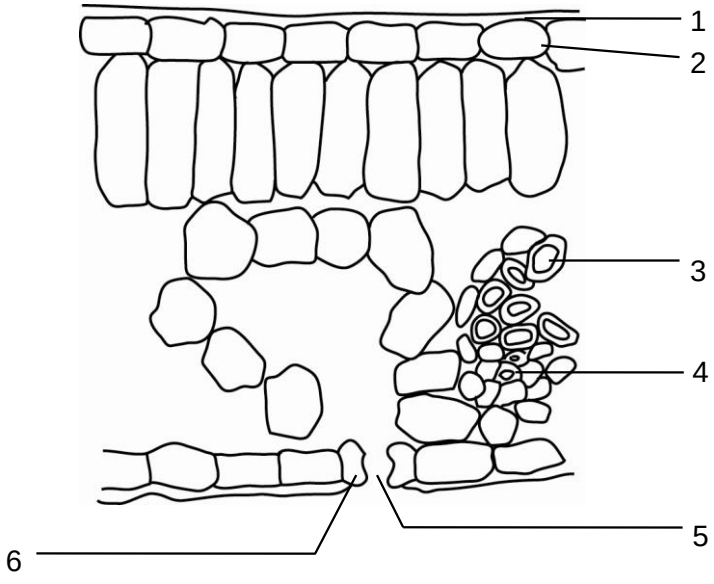
A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0110000037|!|

The diagram below shows a transverse section of a terrestrial dicotyledonous leaf.



Which of the following combinations correctly shows the functions of the structures labelled 1 and 4?

	Structure 1	***Structure 4***
A.	reflects sunlight	transports food
B.	reduces water loss	transports food
C.	reduces water loss	transports water
D.	protects the inner tissues	for mechanical support

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0110000040|!|

Which of the following comparisons between the cell of xylem vessel and sieve tube is ***incorrect***?

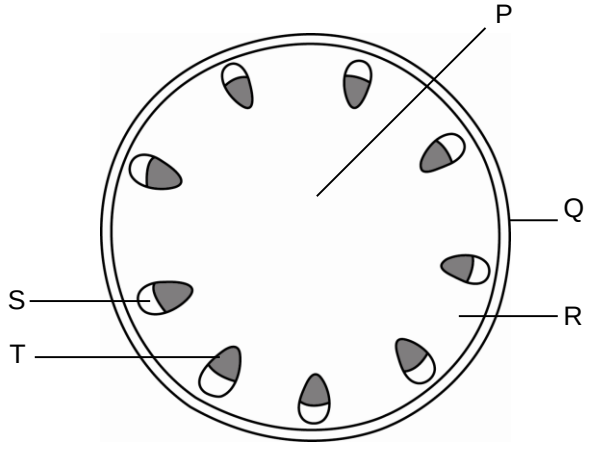
	Cell in xylem vessels	*Cell in sieve tubes*
A.	non-living	living
B.	with lignified cell wall	without lignified cell wall
C.	without a nucleus	with a nucleus
D.	without cytoplasm	with cytoplasm

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0110000041|!|

The diagram below shows a transverse section of a young dicotyledonous stem.



Where does the transport of minerals occur in the stem?

A. P
B. Q
C. S
D. T

##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0110000043|!|

Which of the following environmental conditions will lead to the lowest rate of transpiration?

	Relative humidity	*Wind speed*	*Temperature*
A.	high	high	high
B.	low	low	low
C.	high	low	low
D.	low	high	high

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0110000044|!|

Which of the following environmental conditions will cause the highest rate of transpiration?

	Air	*Light intensity*	*Temperature*
A.	damp	high	low
B.	damp	low	high
C.	dry	high	high
D.	dry	low	low

##C##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0110000045|!|

Young seedlings obtain support mainly from:

(1) turgidity of cortex cells.
(2) turgidity of pith cells.
(3) rigidity of xylem vessels.

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0110000046|!| ★

The supporting tissue, xylem, is located in the periphery of the stem. This can provide

A. mechanical support against bending by wind.
B. better transport of water along the stem.
C. more space for growth in girth.
D. better protection.

##A##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0110000047|!| ★

The supporting tissue, xylem, is located at the centre of the root. This is to provide

A. better protection.
B. better transport of water.
C. more space for growth in girth.
D. better anchorage by acting against the uprooting force of the wind.

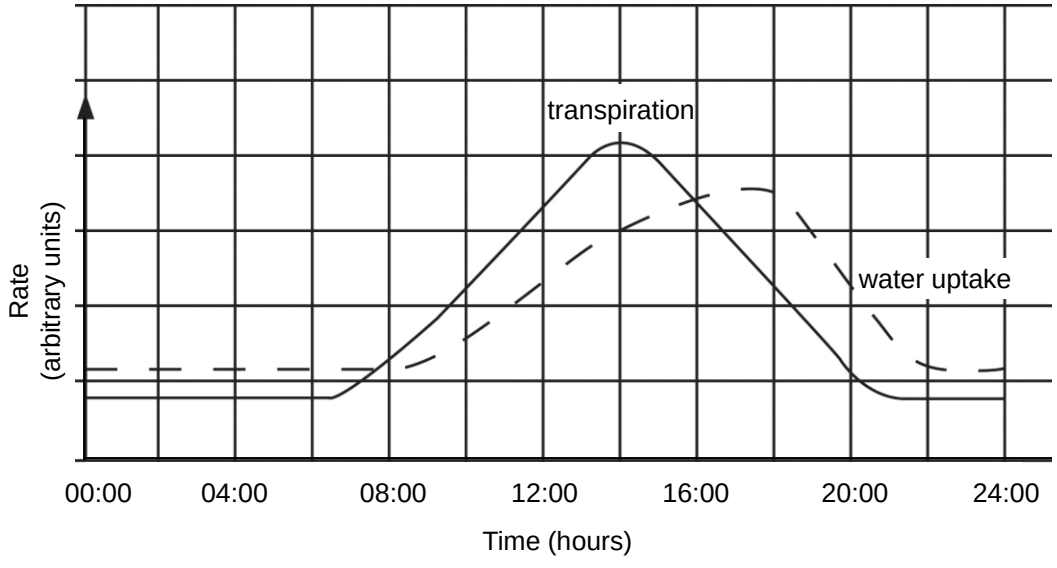
##D##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMB0110000049|!| ★

The graph below shows the rate of transpiration and water uptake by a herbaceous plant during a 24-hour period.



At which of the following time points is the plant most likely to wilt?

A 10:00

B 14:00

C 16:00

D 19:00

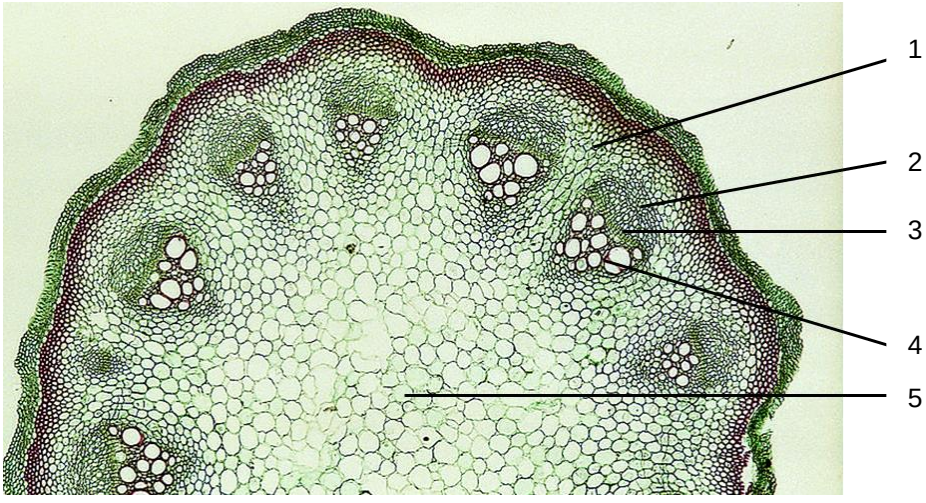
□

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMA0110000050|!| ★★

The photomicrograph below shows a transverse section of part of a young dicotyledonous stem.



(Source: Doc. RNDr. Josef Reischig, CSc.)

Which structures provide support to the plant?

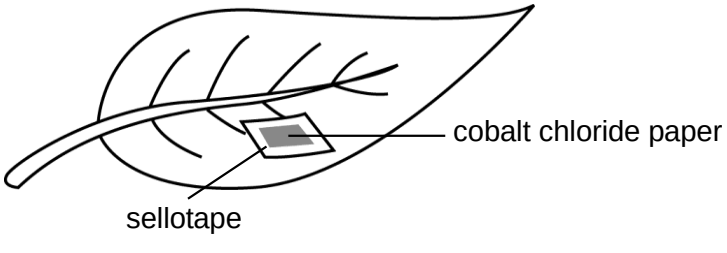
A. 1 and 2 only
B. 1 and 5 only
C. 2 and 4 only
D. 1, 2, 4 and 5 only

##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0110000051|!| ★ SI

An experiment was carried out to investigate water loss through the upper surfaces of leaves in four plants of different species. Dry cobalt chloride papers of the same size were attached onto the upper surfaces of leaves of the four plants as shown below.



The table below shows the results.

Plant	P	Q	R	S
Time taken for the cobalt chloride paper to turn pink (seconds)	1535	167	339	182

Water lost through the upper surfaces of leaves most rapidly in

A. plant P.
B. plant Q.
C. plant R.
D. plant S.

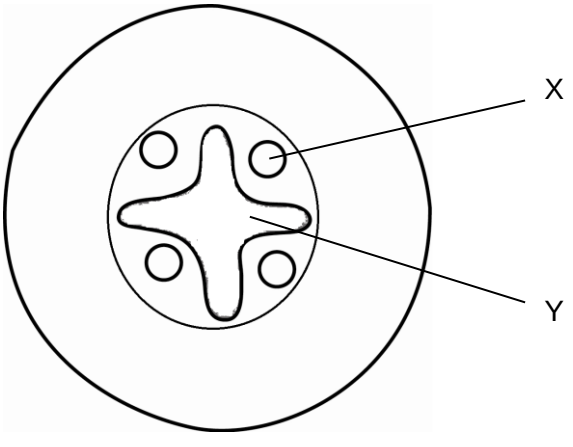
□

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0110000052|!| ★ SI

In an investigation, a herbaceous plant, growing in a coloured nutrient solution, is placed in a well-lit growth chamber through which dry air is being passed slowly. The diagram below shows a section through a part of the plant.



The speeds of movement of the liquids in tissues X and Y are measured. The dry air in the growth chamber is then replaced by humid air and the movements of the liquids change.

Which of the following correctly describes the changes in the movement of the liquids in tissues X and Y?

	Tissue X	*Tissue Y*
A.	little change	increased upward movement
B.	increased downward movement	decreased downward movement
C.	decreased downward movement	decreased upward movement
D.	little change	decreased upward movement

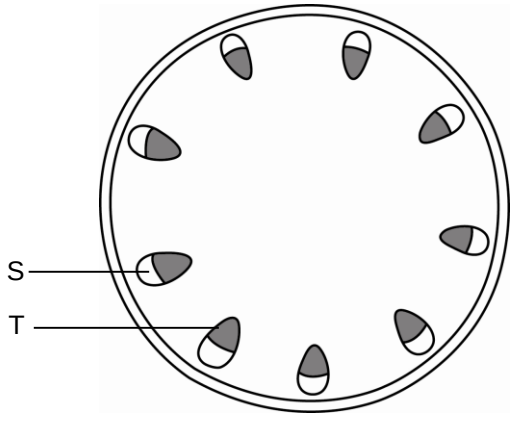
☐

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0110000053|!| ★

The diagram below shows a transverse section of a young dicotyledonous stem.



In which directions do water molecules move in tissue S and in T?

	Tissue S	*Tissue T*
A.	Both up and down	Up only
B.	Down only	Up only
C.	Up only	Down only
D.	Down only	Both up and down

☐

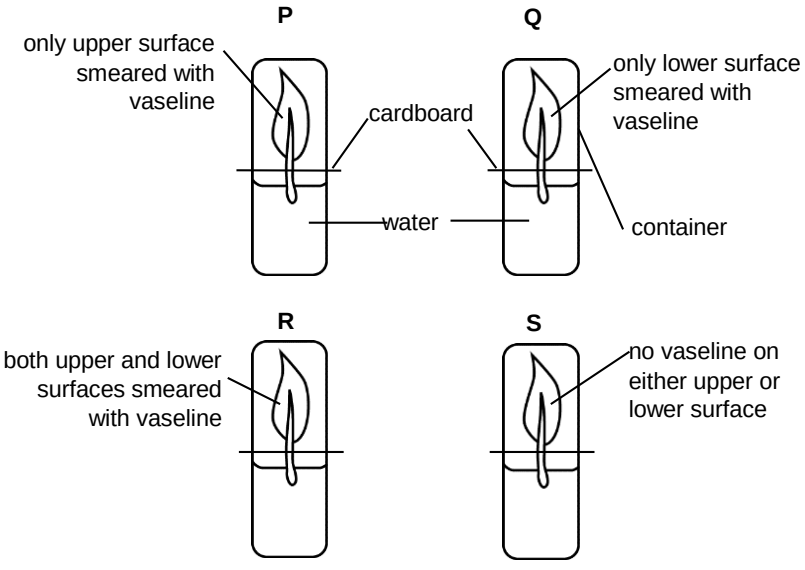
##A##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0110000054|!| ★ SI

Directions: The following two questions refer to the diagrams below, which show the experimental set-up used to investigate the rates of water loss of four similar leaves, detached from a dicotyledonous plant.



After five hours, the inner walls of the containers are either clear or covered by moisture. Which of the following shows the degree of clearness of the wall of container in descending order?

A. R > P > Q > S
B. R > Q > P > S
C. S > P > Q > R
D. S > Q > P > R

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0110000055|!| ★ SI

Which of the following is the reason to explain for the results in the above question?

A. Stomata are present only on the upper surface.
B. Stomatal density is equal on the two surfaces.
C. More stomata are present on the lower surface.
D. More stomata are present on the upper surface.

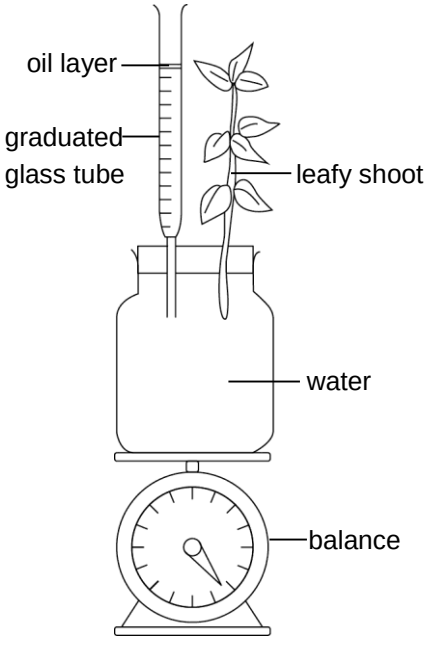
##C##

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Skill assessed	Mark
Applying concepts	1

|!|BCEMA0110000056|!| ★★ SI

Directions: The following two questions refer to the diagram below which shows a weight potometer which was used to study the water balance of a leafy shoot.



The following table shows the results of the experiment after two hours. It is given that the density of water is 1 g /cm^3.

Initial weight of the potometer including the plant	450 g
Final weight of the potometer including the plant	445 g
Initial reading on the graduated glass tube	20 cm^3
Final reading on the graduated glass tube	14.5 cm^3

The rate of transpiration of the leafy shoot was

A. 2.5 g / hour.
B. 2.75 g / hour.
C. 5 g / hour.
D. 5.5 g / hour.

□

##A##

Skill assessed	Mark
Mathematical skills	1

|!|BCEMA0110000057|!| ★★ SI

The amount of water retained by the plant during the experiment was

A. 0.5 g.

B. 5 g.

C. 5.5 g.

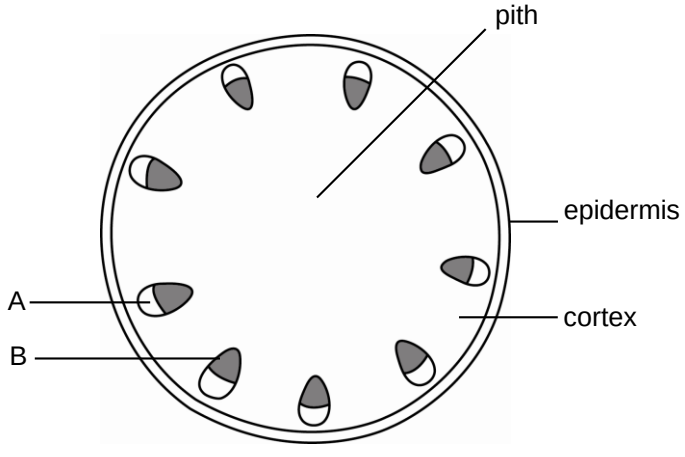
D. 10.5 g.

##A##

Skill assessed	Mark
Mathematical skills	1

|!|BCEMC0110000058|!|

Directions: The following two questions refer to the diagram below which shows a transverse section of the stem.



Which of the following statements about structure A is ***incorrect***?

A. It consists of cells which do not have nuclei.

B. It belongs to the vascular tissues.

C. It consists of cells which do not contain cytoplasm.

D. It helps in the transport of sucrose to the storage organs.

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0110000059|!|

Which of the following is the function of structure B?

A. to absorb water
B. to transport minerals
C. to transport organic nutrients
D. to carry out photosynthesis

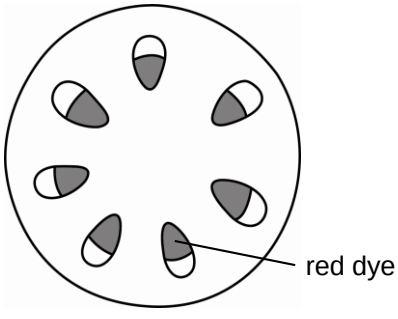
##B##

Skill assessed	Mark
Understanding basic concepts	1

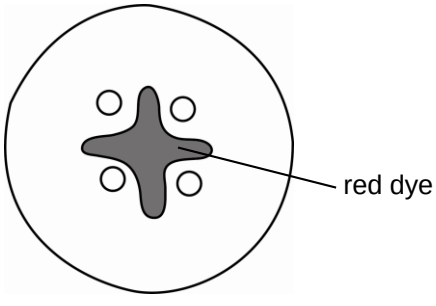
|!|BCEMC0110000060|!|

Directions: The following two questions refer to the diagrams below which show two transverse sections of a plant after the plant's root was immersed in a red dye for five hours.

Region A:



Region B:



Which of the following combinations correctly matches regions A and B to the structures of a plant?

	Region A	***Region B***
A.	root	stem
B.	root	leaf
C.	stem	root
D.	stem	leaf

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0110000061|!|

Which of the following statements about the results is correct?

A. The parts that are stained red contain living cells.

B. The parts that are stained red transports organic nutrients in the plant.

C. The parts that are stained red contain cells with perforated end walls.

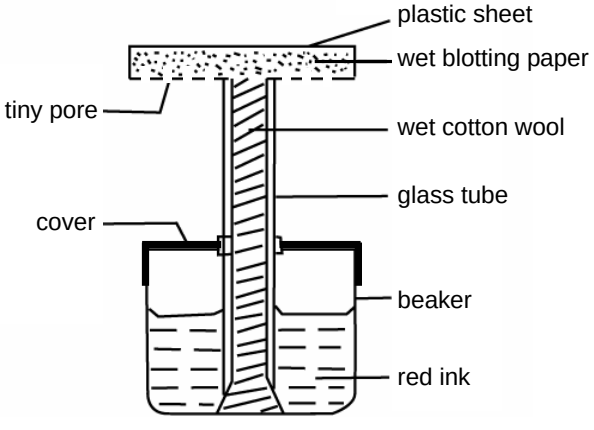
D. The parts that are stained red are responsible for water transport.

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0110000062|!| ★ SI

Directions: The following two questions refer to the diagram below which shows a model which demonstrates the transport of water in a flowering plant.



Which of the following combinations correctly matches the plastic sheet, the tiny pores and the glass tube to their corresponding parts of the plants?

	Plastic sheet	*Tiny pores*	*Glass tube*
A.	cuticle	stomata	xylem
B.	cuticle	guard cells	phloem
C.	upper epidermis	stomata	xylem
D.	upper epidermis	guard cells	phloem

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##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0110000063|!| ★ SI

Which of the following are the possible results of this model after five hours?

(1) The cotton wool is stained red.
(2) The weight of the whole model increases.
(3) The liquid level of the red ink drops.

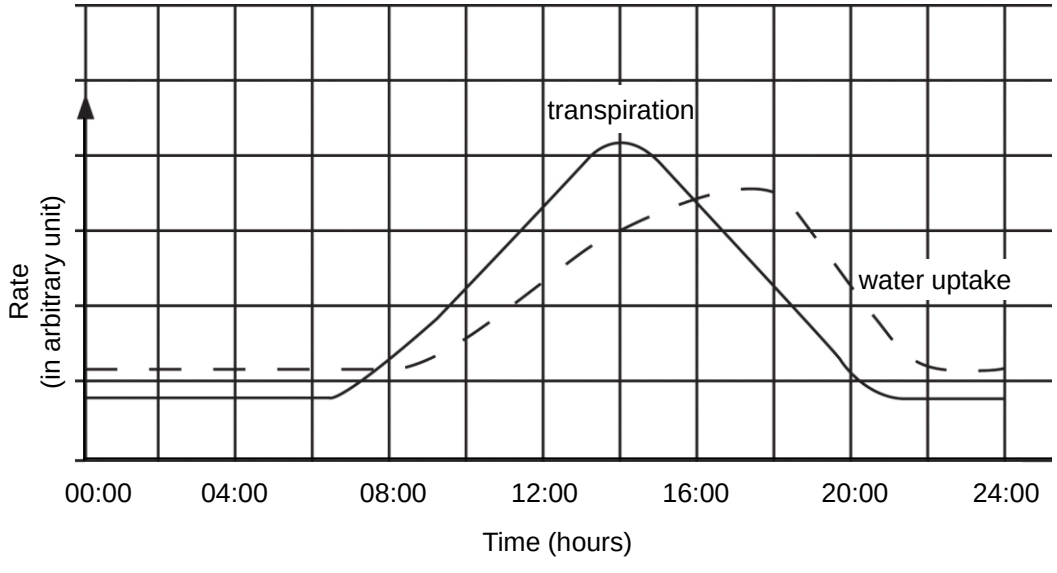
A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0110000064|!| ★

Directions: The following two questions refer to the graph below which shows the rate of transpiration and water uptake by a leafy plant during a 24-hour period.



At which time is the rate of water uptake equal to the rate of transpiration?

A. 00:00 and 07:50
B. 00:00 and 14:00
C. 07:50 and 16:00
D. 14:00 and 17:50

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0110000065|!| ★

Which of the following statements correctly explain the increase in water content of the plant during the time interval from 22:00 to 06:00?

(1) More water is taken up than is lost.
(2) Transpiration stops in darkness.
(3) Stomata are closed at night.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

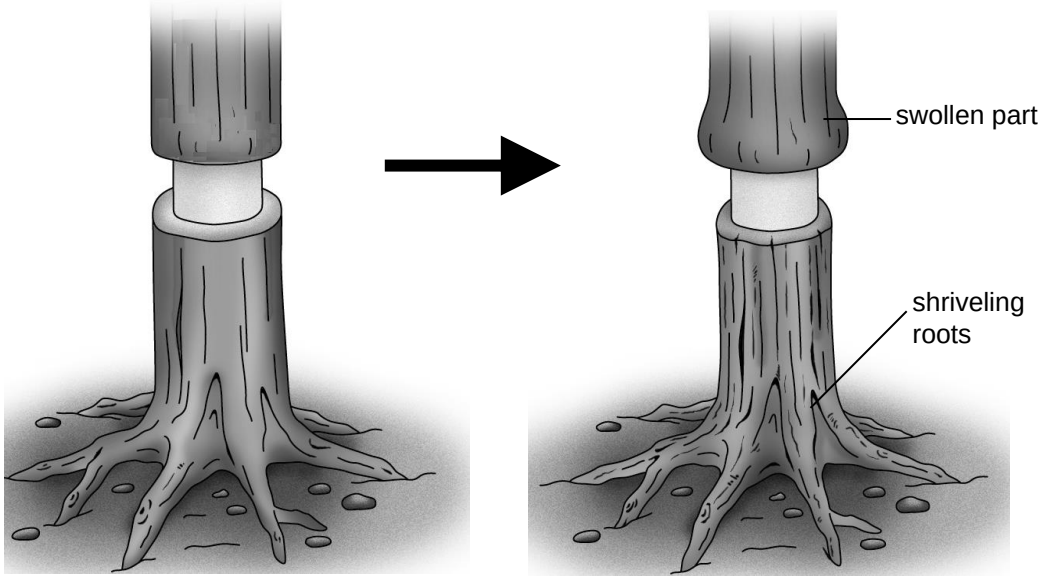
##B##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMB0110000066|!| ★ SI

Directions: The following two questions refer to the ringing experiment used by biologists to study the transport of water and food in plants. The outer ring of the bark of a tree was first removed and the biologists observed the tree a few weeks later. The experimental process and results are shown in the following diagram.



First cut **A few weeks later**

Which of the following can be concluded from this experiment?

A. The outer ring of bark contains tissues that transport food.
B. The outer ring of bark contains tissues that transport water.
C. Food is transported in a single direction from the upper part to the lower part of the tree.
D. Water is transported in a single direction from the lower part to the upper part of the tree.

□

##A##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0110000067|!| ★ SI

After removing the outer ring of the bark, the biologists also cut some of the inner part of the stem. It was found that the leaves wilted in a short period of time. Which of the following ***best*** explains the wilting of the plant?

A. Food is used up as phloem is cut.
B. The tree is subjected to bacterial infections in the cut region.
C. Water evaporates more readily from the cut region.
D. Water cannot be transported to the upper part of the tree.

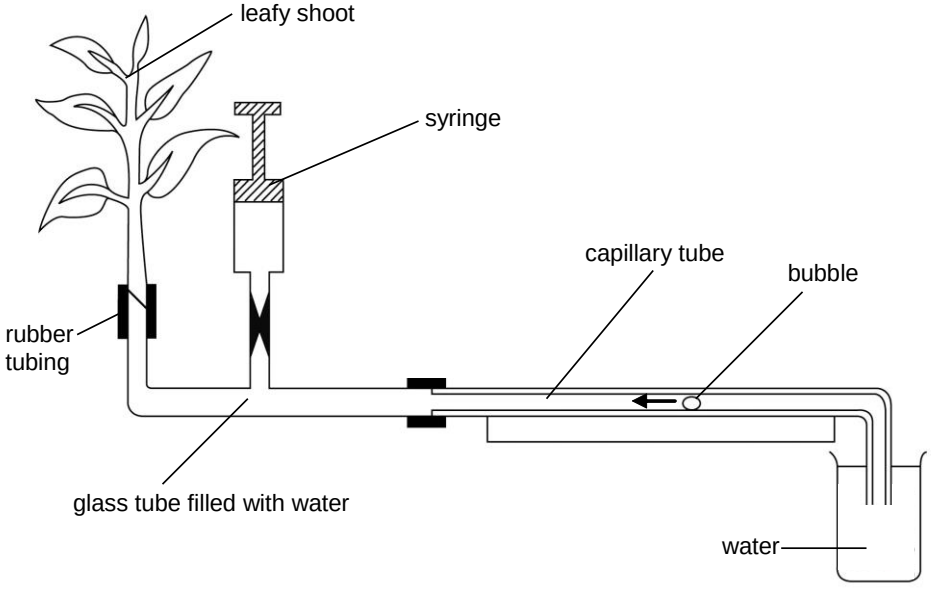
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##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0110000068|!| **SI**

Directions: The following three questions refer to the diagram below which shows a bubble potometer.



Which of the following precautions had to be taken before the experiment?

(1) The whole apparatus should be free of any air bubbles.
(2) The leafy shoot should be cut under water and inserted in the rubber tubing under water.
(3) A layer of oil should be added on the water surface in the beaker.

A. (1) only
B. (2) only
C. (1) and (2) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Practical skills	1

|!|BCEMC0110000069|!| **SI**

The bubble potometer is used to measure the rate of

A. transpiration of the leafy shoot.
B. water loss from the leafy shoot.
C. transport of water in the leafy shoot.
D. water uptake by the leafy shoot.

##D##

Skill assessed	Mark
Practical skills	1

|!|BCEMC0110000070|!| **SI**

The syringe is used to

A. supply more water to the leafy shoot.
B. prevent the entry of air into the apparatus.
C. introduce an air bubble into the capillary tube.
D. return the air bubble to the starting point.

##D##

Skill assessed	Mark
Practical skills	1

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